Joyeta Banerjee

M.Sc. (Engg.) Researcher, ICT · Affective Audio, Human-Centred Design & Emotion-Aware Media

Email: banerjeejoyee3@gmail.com **Phone:** +880 1963 791602 **Location:** Khulna, Bangladesh
Nationality: Bangladeshi **Portfolio:** joyee003.github.io/joyeta-portfolio **LinkedIn:** linkedin.com/in/joyeta-banerjee

ABOUT ME

I am a researcher and designer working on how machines can perceive and respond to human emotion. My M.Sc. thesis at KUET builds a multi-track audio fusion framework that recognises emotion in Bangla recited poetry and instrumental music, and its first results are published as a first-author IEEE conference paper (COMPAS 2025). I also bring 2+ years of industry experience in UX/UI design and product leadership, where I ran user research, prototyped interfaces in Figma, and coordinated cross-functional engineering teams.

> **Research interests:** emotion-aware immersive experiences; affective audio and sonification in AR/VR; user-experience methods for multimodal media; human perception and annotation of emotion; design and evaluation of interactive systems for low-resource cultural contexts.

EDUCATION AND TRAINING

07/2022 – 12/2026 (expected)

M.Sc. (Engg.) in Information and Communication Technology
Khulna University of Engineering & Technology (KUET), Bangladesh
Institute of Information and Communication Technology · Full-time · CGPA 3.50 / 4.00 · 18-credit research thesis
Thesis: *Multi-Track Audio Fusion for Emotion Recognition in Bangla Poetry and Instrumental Music* · Supervisor: Prof. Dr. Mohiuddin Ahmad (Dept. of EEE)

2017 – 2021

B.Sc. (Engg.) in Computer Science and Engineering
Northern University of Business and Technology Khulna (NUBTK), Bangladesh
CGPA 3.24 / 4.00 · HSC (Science) GPA 4.00/5.00, SSC (Science) GPA 4.88/5.00, Satkhira

RESEARCH EXPERIENCE

07/2022 – Present

M.Sc. Thesis Researcher: Affective Audio & Emotion Recognition
KUET, Khulna · Supervisor: Prof. Dr. Mohiuddin Ahmad

- Published the first results as **first author at IEEE COMPAS 2025**: MFCC, spectral and temporal features from Bangla song recordings, combined by feature-level fusion and classified with a CNN, compared against SVM, Random Forest and CatBoost.
- Designing a **multi-track audio fusion framework** that combines time-aligned vocal (prosodic) and instrumental (timbral, rhythmic, harmonic) cues to recognise six emotions (Calm, Sad, Reflective, Devotional, Peaceful, Neutral) more accurately than single-track systems.
- Building the first Bangla corpus that pairs **recited poetry with instrumental music**: a studio-recorded seed set (narration plus five instrumental versions of a song on sarod, flute, harmonica, guitar and piano), scaling to ~170 recordings and ~15,000 clips.
- Designed the **human annotation protocol**: three independent raters, majority vote, Fleiss' and Cohen's $\kappa \geq 0.61$ with re-annotation of low-agreement clips, plus supplementary **valence–arousal** ratings to study how listeners perceive emotion.
- Developing a **leakage-safe pipeline** with recording-level splits, DTW time alignment, stem separation, and training-only augmentation (time-stretch, pitch-shift, noise, spectrogram masking). The pipeline compares concatenation, weighted fusion and statistical aggregation of MFCC, spectral and temporal features across classical ML and CNN models.

PUBLICATIONS

10/2025 Peer-reviewed

J. Banerjee, M. Ahmad, and T. Tabassum, "Multi-Track Audio Fusion for Emotion Recognition Using Machine and Deep Learning Techniques," in *Proc. 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS 2025)*, Kushtia, Bangladesh, Oct. 2025. ieeexplore.ieee.org/document/11381614

In preparation

J. Banerjee and M. Ahmad, "Multi-Track Audio Fusion for Emotion Recognition in Bangla Poetry and Instrumental Music": extended study with an expanded speech–music corpus, human annotation and leakage-safe recording-level evaluation.

PROFESSIONAL EXPERIENCE

10/2024 – 07/2026
Hybrid

Project Manager (UI/UX, Frontend & Backend Projects)
Creatibuzz LLC

- Led discovery sessions and stakeholder interviews and turned user needs into PRDs, user stories and technical documentation.
- Coordinated design, frontend and backend teams through Agile/Scrum sprints, milestone tracking and risk mitigation for international clients.

09/2024 – 02/2026
Hybrid

Technical Project Coordinator
Akij iBOS Limited

- Managed ERP and digital products end to end, from discovery to delivery. Ran market research and competitor analysis to define requirements and technical feasibility.
- Audited user-centred design layouts and frontend features against stakeholder goals, and acted as the link between international clients and engineering teams.

09/2024 – 11/2024
Remote

Education Project Manager
ishQool
Coordinated content, design and technical updates for feature rollouts on a learning platform.

05/2024 – 02/2025
Remote

Technical Content Writer (Fileion) · Content Writing Intern (YSSE)
Translated complex software features into user guides, FAQs and knowledge-base articles. Wrote and edited research-informed, SEO-optimised content.

12/2020 – 03/2021
Onsite

Frontend Web Developer
Zaman Tech

- Implemented user-centred UI features with designers and backend engineers and gave technical-feasibility input to product decisions.

DESIGN PROJECTS

2025 – 2026

Product & UX Case Studies: Shuddhi, Shonchoy, Moodbrew (+2)
End-to-end UX in Figma: user interviews, personas, journey maps, design systems, high-fidelity interactive prototypes, and developer handoff. joyee003.github.io/joyeta-portfolio

SKILLS

Research methods	Corpus design & data collection · human annotation and inter-rater reliability (Fleiss'/Cohen's κ) · valence–arousal ratings · experimental design · statistical testing · user interviews · usability evaluation
Audio & ML	Audio signal processing (STFT, MFCC, spectral and temporal features) · data augmentation · feature-level fusion · classical ML (SVM, Random Forest, CatBoost) and CNN classification · cross-validation, ROC-AUC, macro-F1 · **Python, librosa, scikit-learn, PyTorch**
Design & prototyping	Figma (design systems, variables, auto layout) · wireframing · interactive prototyping · UX writing · HTML5, CSS3, Bootstrap · Git & GitHub
Product & collaboration	Agile/Scrum · Jira · PRDs and user stories · RICE and MoSCoW prioritisation · leading cross-functional teams · international stakeholder communication

LANGUAGE SKILLS

Mother tongue: Bengali

English test: IELTS Academic, scheduled December 2026

Other language	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	C1	C1	C1	C1	C1

Levels: A1/A2 basic user · B1/B2 independent user · C1/C2 proficient user (CEFR, self-assessed)

CERTIFICATIONS & CONFERENCES

IEEE COMPAS 2025: First-Author Paper

Foundations of User Experience (UX) Design (Coursera)

Intro to UX Writing · Git & GitHub · Responsive Web Design (Udemy)

Technical Product Management (LinkedIn)

Figma UI/UX Design Essentials (Udemy)

Mobile Game & Application Skill Development (ICT Ministry, BD)

REFERENCES

Prof. Dr. Mohiuddin Ahmad
Professor, Dept. of EEE, KUET · Thesis Supervisor
ahmad@eee.kuet.ac.bd

Tahmida Tabassum
IICT, KUET · Co-author
tabassum@iict.kuet.ac.bd